

* $\left| \frac{x - \bar{x}}{\bar{x}} \right| < \frac{1}{2} \bar{x} 10^{-d}$ eşitsizliği sağlanacak şekilde en büyük d pozitif tam sayısı mevcut sa x sayısı $\bar{x}$ sayısına d anlamlı basamak yakındır denir.

örnek:

$$x = 3,141592 \quad \bar{x} = 3,14$$

$$\left| \frac{x - \bar{x}}{\bar{x}} \right| = 0,000507 = \frac{1}{2} \times 10^{-3} \quad 3 \text{ anlamlı basamak yakındır.}$$

örnek:

$$x = 1.000.000 \quad \bar{x} = 999.996$$

$$\left| \frac{x - \bar{x}}{\bar{x}} \right| = 0,000004 < 0,000005 = \frac{1}{2} \times 10^{-5}$$

örnek:

$$x = 0,00012 \quad \bar{x} = 0,000009$$

$$\left| \frac{x - \bar{x}}{\bar{x}} \right| = 0,25 < 0,5 = \frac{1}{2} \times 10^{-9}$$

Truncation Error: (Kesme Hatanı). The notion of truncation error usually refers to errors introduced when a more complicated mathematical expression is "replaced" with a more elementary formula. This terminology originates from the technique of replacing a complicated function with a truncated Taylor series.

Sayısal işlemlerde kullanılan sonsuz uzunlukta bir serinin belli bir yerde kesilip geri kalan terimlerin atılması sık sık yapılmaktadır. Buradan ortaya çıkan yanlışı kesme yanlışıdır.

Örneğin; $e^x = 1 + x + \frac{x^2}{2!} + \dots + \frac{x^n}{n!} + \dots$

$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} + \dots + \frac{(-1)^n x^{2n+1}}{(2n+1)!} + \dots$

Ne kadar sonsuz serilerdir.

The Asymptotic Symbols:

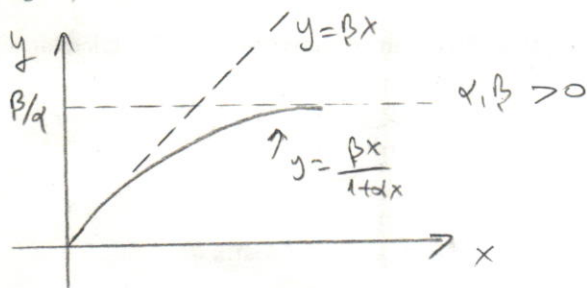
The $\sim$ Symbol, The Big "O" Symbol and the Little "o" symbol.

Definition: If $\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = 1 \Rightarrow$ we write $f(x) \sim g(x)$ as $x \rightarrow x_0$
 $f(x)$ and $g(x)$ are said to be equivalent

(see overleaf)

Örneği: $f(x) = \frac{\beta x}{1+\alpha x}$ ise a) $f(x) \sim \beta x$ as $x \rightarrow 0$

$\lim_{x \rightarrow 0} \frac{f(x)}{g(x)} = \lim_{x \rightarrow 0} \frac{1}{1+\alpha x} = 1$ then $f(x) \sim \beta x$ as $x \rightarrow 0$



b) $f(x) \sim \frac{\beta}{\alpha}$ as $x \rightarrow \infty$

since

$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \lim_{x \rightarrow \infty} \frac{\alpha x}{1+\alpha x} = 1$

$g(x) = \frac{\beta x}{1+\alpha x} \Rightarrow$ a) $g(x) \sim \beta x$ as $x \rightarrow 0$

b) $g(x) \sim \frac{\beta}{\alpha}$ as $x \rightarrow \infty$

Ornek: $\sin x = x - \frac{x^3}{3!} + \dots$ (yikinci mertebeden itibaren)
 $\sin x \sim x$ as $x \rightarrow 0$ since $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$

we can show that $\cos x \sim 1$ as $x \rightarrow 0$ $\lim_{x \rightarrow 0} \frac{\cos x}{1} = 1$
 $\log x \sim x$ as $x \rightarrow 0$ $\lim_{x \rightarrow 0} \frac{\log x}{x} = 1$
 $\exp x \sim 1$ as $x \rightarrow 0$
 $\ln(1+x) \sim x$ as $x \rightarrow 0$

we can improve our correspondence by computing more terms in the Taylor series and using them in our relations. For instance,

$\sin x \sim x - \frac{x^3}{3!}$ as $x \rightarrow 0$

or $\frac{\sin x}{x - \frac{x^3}{3!}} \rightarrow 1$ as $x \rightarrow 0$

Definition: we write $f(x) = O[g(x)]$ as $x \rightarrow x_0$

If $\lim_{x \rightarrow x_0} \frac{|f(x)|}{|g(x)|} \leq K$, $K > 0$ constant.

$|f(x)| \leq K|g(x)|$

Öyle bir K sayısı ve x_0 -in bir komşuluğu vardır ki

Definition: we write $f(x) = o[g(x)]$ as $x \rightarrow x_0$

If $\lim_{x \rightarrow x_0} \frac{|f(x)|}{|g(x)|} = 0$

$x^2 = o(x)$ as $x \rightarrow 0$, $\frac{\log x}{x} = o(1)$ as $x \rightarrow \infty$
 $\frac{\sin x}{x} = o(1)$ as $x \rightarrow \infty$

Thus, these symbols characterize different states of our knowledge of the asymptotic behaviour of a function:

- $\sim$ = we know the most
- o = we know less
- O = we know the least.

Ornek: we know that $\sin x$ is bounded for all x . So we can

write $\sin x = O(1)$ for all x . $\left| \frac{\sin x}{1} \right| \leq 1$

$f(x) = \frac{1}{x} \Rightarrow f(x) = O(x)$ as $x \rightarrow \infty$ since $|f(x)|/|x| \leq k$ as $x \rightarrow \infty$
 $f(x) = O(1)$ as $x \rightarrow 0$ since $|f(x)|/1 \leq k$ as $x \rightarrow 0$

$\sin x = x + O(x^3)$ as $x \rightarrow 0$
 $\sin x = x + O(x^3)$ as $x \rightarrow 0$

$\frac{\sin x}{x} = o(1)$ as $x \rightarrow \infty$
 $\frac{\log x}{x} = o(1)$ as $x \rightarrow \infty$

$O(h^n)$ order of Approximation:

Definition: Assume that $f(h)$ is approximated by the function $p(h)$ and that there exists a real constant $M > 0$ and a positive integer n so that

$$\frac{|f(h) - p(h)|}{|h|^n} \leq M \text{ for sufficiently small } h. (*)$$

we say that $p(h)$ approximates $f(h)$ with order of approximation $O(h^n)$ and write $f(h) = p(h) + O(h^n)$.

when relation (*) is rewritten in the form $|f(h) - p(h)| \leq M|h|^n$ we see that the notation $O(h^n)$ stands in place of the error bound $M|h|^n$.

Theorem: Assume that $f(h) = p(h) + O(h^n)$, $g(h) = q(h) + O(h^m)$ and $r = \min\{m, n\}$. Then

$$f(h) + g(h) = p(h) + q(h) + O(h^r)$$

$$f(h)g(h) = p(h)q(h) + O(h^r)$$

$$\frac{f(h)}{g(h)} = \frac{p(h)}{q(h)} + O(h^r) \text{ provided that } g(h) \neq 0 \text{ and } q(h) \neq 0.$$

Taylor's Theorem: Assume that $f \in C^{n+1}[a, b]$. If both x_0 and $x_0 + h$ lie in $[a, b]$, then

$$f(x_0 + h) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} h^k + O(h^{n+1}), \quad \frac{f^{(n+1)}(c)}{(n+1)!} h^{n+1} \approx M h^{n+1} \approx O(h^{n+1})$$

$$i) \quad O(h^p) + O(h^p) = O(h^p)$$

$$ii) \quad O(h^p) + O(h^q) = O(h^r), \quad r = \min\{p, q\}$$

$$iii) \quad O(h^p) O(h^q) = O(h^s), \quad s = p + q$$

Ex: ex 1:

$$e^h = 1 + h + \frac{h^2}{2!} + \frac{h^3}{3!} + O(h^4) \quad \text{we} \quad \cosh h = 1 + \frac{h^2}{2!} + \frac{h^4}{4!} + O(h^6)$$

determine the order of approximation for their sum and product.

$$\text{Solution: } e^h + \cosh h = 2 + h + \frac{h^3}{3!} + O(h^4) + \frac{h^4}{4!} + O(h^6)$$

$$O(h^4) + \frac{h^4}{4!} = O(h^4)$$

$$O(h^4) O(h^6) = O(h^4)$$

$$e^h + \cosh h = 2 + h + \frac{h^3}{3!} + O(h^4)$$

$$e^h \cos h = \left(1 + h + \frac{h^2}{2!} + \frac{h^3}{3!}\right) \left(1 - \frac{h^2}{2!} + \frac{h^4}{4!}\right) + \left(1 + h + \frac{h^2}{2!} + \frac{h^3}{3!}\right) O(h^6) \\ + \left(1 - \frac{h^2}{2!} + \frac{h^4}{4!}\right) O(h^4) + O(h^4) O(h^6)$$

$$e^h \cos h = 1 + h - \frac{h^3}{3} - \frac{5h^4}{24} - \frac{h^5}{24} + \frac{h^6}{48} + \frac{h^7}{144} + O(h^6) + O(h^4) + \underbrace{O(h^4) O(h^6)}_{O(h^{10})}$$

$$\boxed{e^h \cos h = 1 + h - \frac{h^3}{3} + O(h^4)}$$

order of convergence of a sequence:

Definition: Suppose that $\lim_{n \rightarrow \infty} x_n = x$ and $\{r_n\}_{n=1}^{\infty}$ is a sequence with

$\lim_{n \rightarrow \infty} r_n = 0$. We say that $\{x_n\}_{n=1}^{\infty}$ converges to x with the order of convergence $O(r_n)$ if there exists a constant $K > 0$ such that

$$\frac{|x_n - x|}{|r_n|} \leq K \text{ for } n \text{ sufficiently large.}$$

This is indicated by writing $x_n = x + O(r_n)$ or $x_n \rightarrow x$ with order of convergence $O(r_n)$.

Ex: Let $x_n = \cos(n)/n^2$ and $r_n = \frac{1}{n^2}$, then $\lim_{n \rightarrow \infty} x_n = 0$ with a rate of convergence $O(1/n^2)$.

Solution: This follows immediately from the relation

$$\frac{|\cos(n)/n^2|}{|1/n^2|} = |\cos(n)| \leq 1 \text{ for all } n.$$

Örnek:

(1)

- 1) a) $x - 0.2 \sin x - 0.5 = 0$ Denkleminin $[0,1]$ aralığındaki kökü için Aralık yarılama metodu ile 3 iterasyon yapınız (her iterasyonda aralığı belirtiniz)
b) Bu kökü 10^{-4} doğrulukta hesaplamak için kaç iterasyon yapılmalıdır.

a) $[0,1]$ $f(0) = -0.5 < 0$ $f(1) = 0.3317 > 0$

1. iterasyon: $c_1 = \frac{0+1}{2} = 0.5$ $f(0.5) = -0.0958 < 0$, kök $[0.5, 1]$ aralığında

2. iterasyon: $c_2 = \frac{0.5+1}{2} = 0.75$ $f(0.75) = 0.1136 > 0$, kök $[0.5, 0.75]$ " "

3. iterasyon: $c_3 = \frac{0.75+0.5}{2} = 0.625$ $f(0.625) = 0.0079 > 0$, kök $[0.5, 0.625]$ " "

b) $n = \tilde{n} \geq \frac{\log(b_0 - a_0) + k}{\log 2} \Rightarrow n = \tilde{n} \geq \frac{4}{\log 2} = 13.2 \Rightarrow n = 14$ iterasyon.

- 2) Aralık yarılama metodu kullanarak $\sin(2x) - e^{-x} - 1 = 0$ Denkleminin $[0.5, 1]$ aralığındaki kökü için üç iterasyon yapınız.

$f(0.5) = 0.235 > 0$ $f(1) = -0.09 < 0$

$c_0 = \frac{1+0.5}{2} = 0.75$ $f(0.75) = 0.219 > 0$ kök $[0.75, 1]$ aralığında

$c_1 = \frac{1+0.75}{2} = 0.875$ $f(0.875) = 0.101 > 0$ kök $[0.875, 1]$ " "

$c_2 = \frac{1+0.875}{2} = 0.937$ $f(0.937) = 0.016 > 0$ kök $[0.937, 1]$ aralığında

Örnek: $3x - \cos x - 1 = 0$ denkleminin kökü için $x_0 = 0$ ve $x_1 = 1$ değerlerini kullanarak Regula Falsi yöntemi ile 3 iterasyon yapınız. (Her iterasyonda iterasyon formülünü ve aralığı belirtiniz)

$$f(x) = 3x - \cos x - 1 \quad f(0) = -2 < 0, \quad f(1) = 1.4596 > 0$$

$$[x_0, x_1] = [0, 1]$$

$$x_2 = x_0 - \frac{x_1 - x_0}{f(x_1) - f(x_0)} \cdot f(x_0) \Rightarrow x_2 = 0 - \frac{1}{1.4596 + 2} \cdot (-2) = 0.5781$$

$$f(0.5781) = -0.1032 > 0 \quad \text{ve} \quad [x_2, x_1] = [0.5781, 1]$$

$$x_3 = x_2 - \frac{x_1 - x_2}{f(x_1) - f(x_2)} \cdot f(x_2) \Rightarrow x_3 = 0.5781 - \frac{1 - 0.5781}{1.4596 - (-0.1032)} \cdot (-0.1032)$$

$$x_3 = 0.6059$$

$$f(0.6059) = -0.0042 < 0 \quad [x_3, x_1] = [0.6059, 1]$$

$$x_4 = x_3 - \frac{x_1 - x_3}{f(x_1) - f(x_3)} \cdot f(x_3) \Rightarrow x_4 = 0.6059 - \frac{1 - 0.6059}{1.4596 - (-0.0042)} \cdot (-0.0042)$$

$$x_4 = 0.6070$$

$$f(0.6070) = -0.0036 < 0 \quad [x_4, x_1] = [0.6070, 1]$$

Örnek: $f(x) = 2 + \cos(e^x - 2) - e^x \quad x_0 = 0.5, \quad x_1 = 1.5$

Aynı sorun.

Örnek: $3x - \cos x - 1 = 0$ denkleminin kökünü için $x_0 = 0$ ve $x_1 = 1$ değerlerini kullanarak Regula Falsi metodu ile 3 iterasyon yapınız. (Her iterasyonda iterasyon formülünü ne anladığınızı belirtiniz)

$$f(x) = 3x - \cos x - 1 \quad f(0) = -2 < 0, \quad f(1) = 1.4596 > 0$$

$$[x_0, x_1] = [0, 1]$$

$$x_2 = x_0 - \frac{x_1 - x_0}{f(x_1) - f(x_0)} \cdot f(x_0) \Rightarrow x_2 = 0 - \frac{1}{1.4596 + 2} \cdot (-2) = 0.5781$$

$$f(0.5781) = -0.1032 > 0 \quad \text{kök} \quad [x_2, x_1] = [0.5781, 1]$$

$$x_3 = x_2 - \frac{x_1 - x_2}{f(x_1) - f(x_2)} \cdot f(x_2) \Rightarrow x_3 = 0.5781 - \frac{1 - 0.5781}{1.4596 - (-1.1032)} \cdot (-0.1032)$$

$$x_3 = 0.6059$$

$$f(0.6059) = -0.0042 < 0 \quad [x_3, x_1] = \begin{matrix} 0.6059 & 1 \\ x_3 & x_1 \end{matrix}$$

$$x_4 = x_3 - \frac{x_1 - x_3}{f(x_1) - f(x_3)} \cdot f(x_3) \Rightarrow x_4 = 0.6059 - \frac{1 - 0.6059}{1.4596 - (-0.0042)} \cdot (-0.0042)$$

$$x_4 = 0.6070$$

$$f(0.6070) = -0.0036 < 0 \quad [x_4, x_1] = [0.6070, 1]$$

Örnek: $f(x) = 2 + \cos(e^x - 2) - e^x \quad x_0 = 0.5, \quad x_1 = 1.5$
Aynı soru.